

JSP3: Metaphor quality control

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1 Introduction

This report evaluates several machine learning models for distinguishing between metaphoric and metonymic semantic shifts. The experiments were performed on the ChainNet dataset, which contains 13,616 annotated examples consisting of 7,500 metaphors and 6,116 metonymies.

Several classifiers were compared using identical feature representations derived from WordNet and sentence embeddings. Model performance was evaluated using 10-fold cross-validation. Besides classification accuracy, we also analyzed precision, recall, F1-score, calibration curves, and confusion matrices.

2 Background

The task is formulated as a binary classification problem. Given two WordNet senses connected by a semantic shift, the classifier predicts whether the relation represents a metaphor or a metonymy.

WordNet provides lexical-semantic relations between concepts, while Sentence Transformers are used to obtain dense semantic representations of synset definitions.

Resources

Since both metaphor and metonymy represent semantic shifts between word senses, the ChainNet resource was used. We specifically employed the files **chainnet_metaphor.json** and **chainnet_metonymy.json**, which provide the word form together with the source and target WordNet senses.

3 Approach

Training was provided in python and code was located in the file `train_mqc.py`. First we needed to load input files.

For every pair of WordNet synsets, a feature vector was extracted. The features combine semantic similarity obtained from sentence embeddings with structural information derived from the WordNet hierarchy.

Features
cosine similarity
path similarity
wup similarity
jaccard hypernyms
semantic gap
hypernym overlap
depth difference
semantic distance
lch depth
transition_lexfile
from_pos
from_lexfile
definition similarity
from_sense definition
from_sense embedding

4 Results

Model Accuracy

Table 1: Accuracy per model

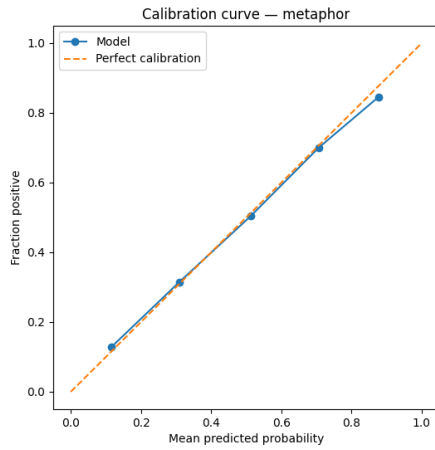
Model	Train	Test
RandomForestClassifier	99.75%	70.89%
LogisticRegression	92.80%	69.73%
XGBClassifier	70.74%	69.13%
KNeighborsClassifier	99.75%	66.38%
GaussianNB	81.19%	59.65

F1 score

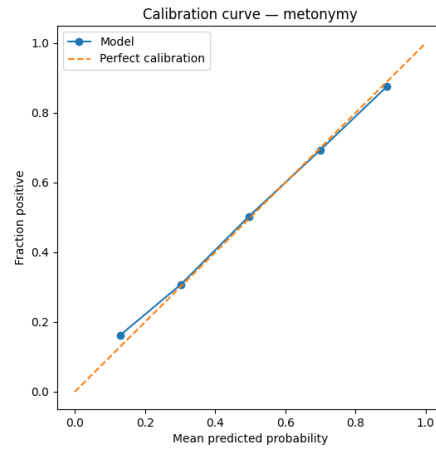
Table 2: Test F1

Model	precision	recall	F1
RandomForestClassifier	70.95%	69.80%	69.93%
LogisticRegression	69.42%	69.23%	69.29%
XGBClassifier	69.37%	67.78%	67.82%
KNeighborsClassifier	66.00%	65.67%	65.73%
GaussianNB	64.30%	61.90%	58.60%

Per_prediction confidence

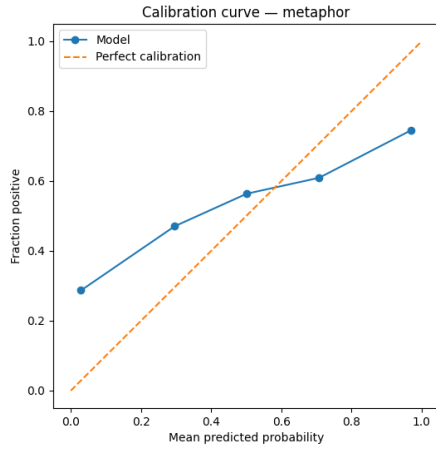


(a) Metaphor confidence

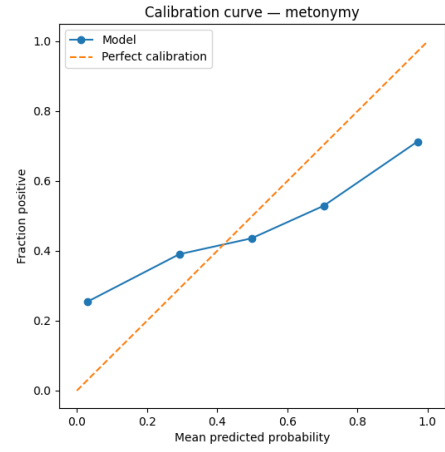


(b) Metonymy confidence

Figure 1: Calibration curves for model Random Forest.

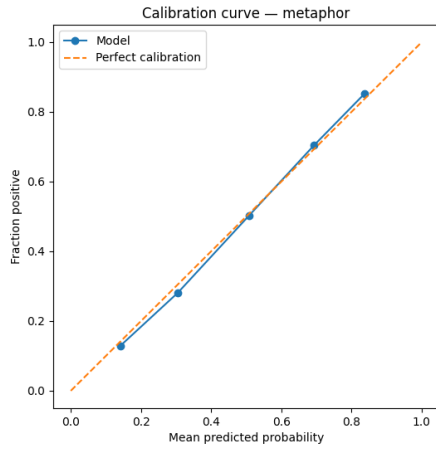


(a) Metaphor confidence

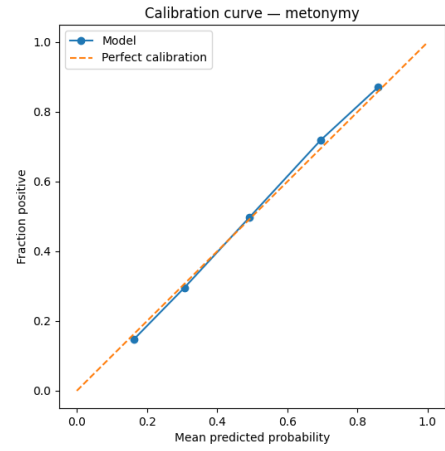


(b) Metonymy confidence

Figure 2: Calibration curves for model L. Regression

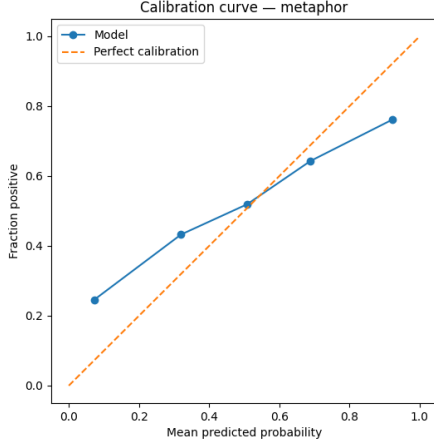


(a) Metaphor confidence

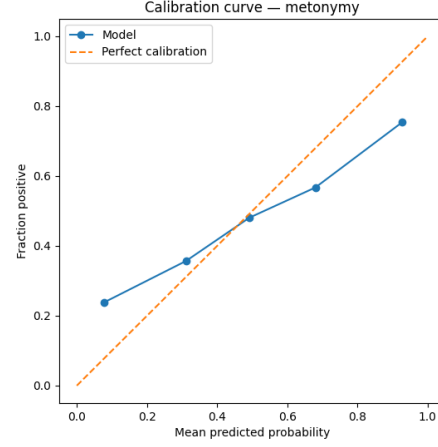


(b) Metonymy confidence

Figure 3: Calibration curves for model R. XGBClassifier.

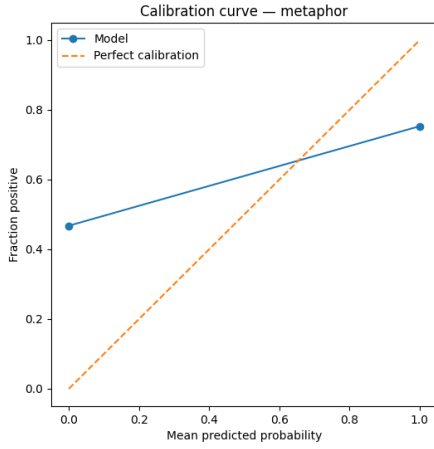


(a) Metaphor confidence

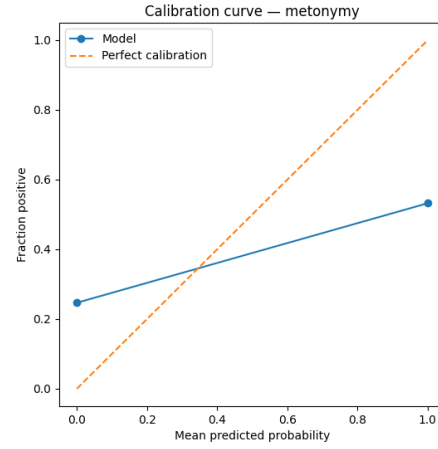


(b) Metonymy confidence

Figure 4: Calibration curves for model KNClassifier.



(a) Metaphor confidence



(b) Metonymy confidence

Figure 5: Calibration curves for model Naive Bayes.

The calibration curves indicate that Random Forest and Logistic Regression produce reasonably calibrated probability estimates. The curves remain close to the ideal diagonal, although confidence tends to be slightly overestimated for higher predicted probabilities. Gaussian Naive Bayes exhibits noticeably poorer calibration.

Error analysis

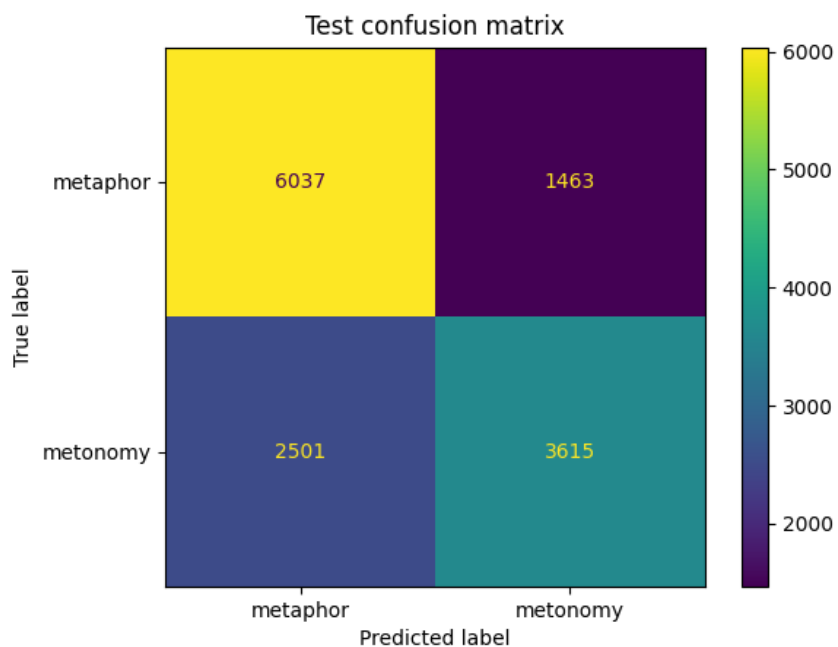


Figure 6: Confusion matrix for model R. Forest

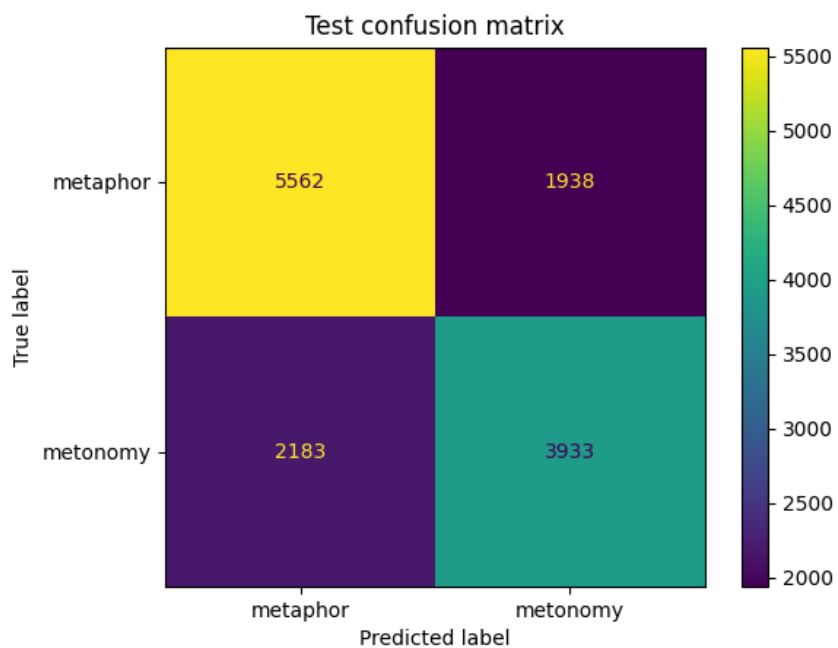


Figure 7: Confusion matrix for model Log. Regression .

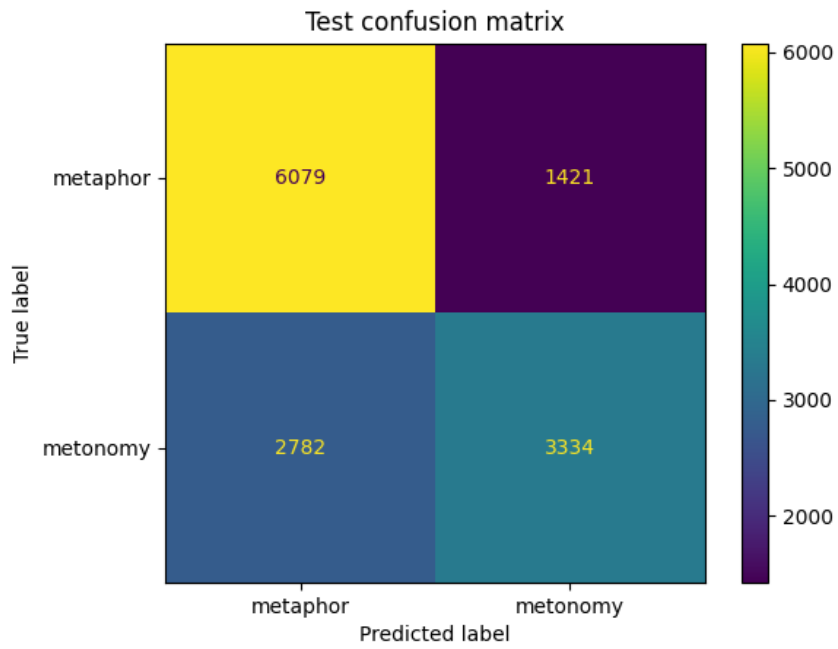


Figure 8: Confusion matrix for model XGBClassifier.

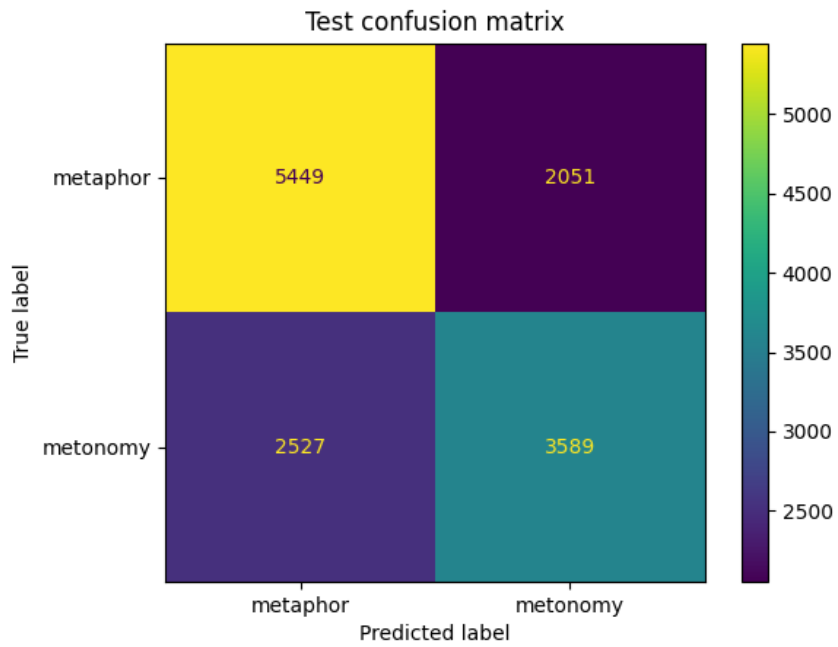


Figure 9: Confusion matrix for model KNNClassifier.

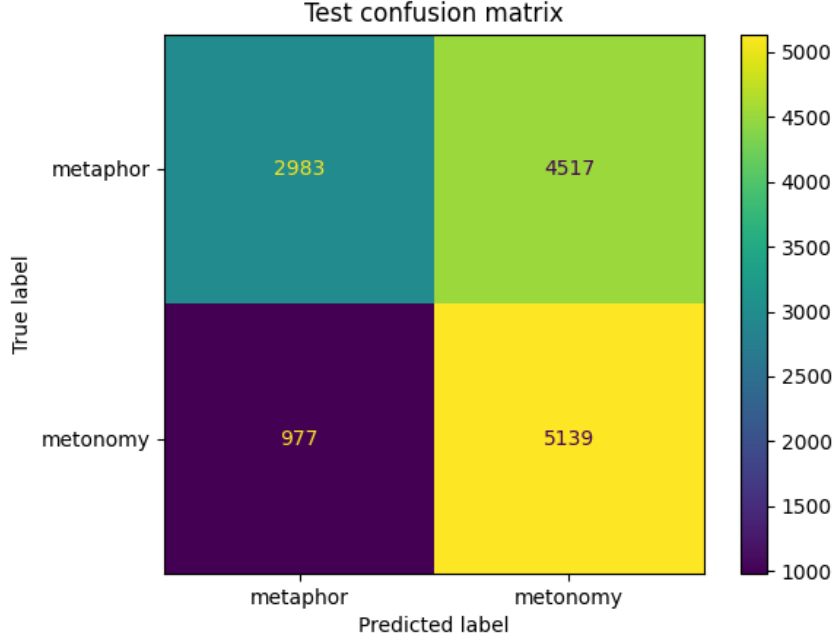


Figure 10: Confusion matrix for model Naive Bayes.

Random Forest achieved the highest overall performance, reaching an average test accuracy of 70.89% and the highest macro F1-score (69.93%). Logistic Regression achieved comparable performance, while XGBoost produced similar but slightly worse results. KNN and Gaussian Naive Bayes performed considerably worse.

5 Discussion

The experimental results suggest that distinguishing metaphor from metonymy using only semantic similarity and WordNet-based structural features is a challenging task. Although Random Forest achieved almost 71% accuracy, the improvement over Logistic Regression is relatively small.

The large gap between training and testing accuracy for Random Forest and KNN indicates substantial overfitting. Logistic Regression generalized considerably better despite achieving slightly lower training accuracy.

The error analysis further demonstrates that many metaphors and metonymies share similar semantic structures, making them difficult to separate using lexical resources alone. Future work could include contextual language models, graph neural networks over WordNet, or additional linguistic features

describing semantic transformations.

6 Conclusions

This work compared several machine learning approaches for metaphor quality control using semantic features extracted from WordNet and sentence embeddings.

Random Forest achieved the best overall performance with approximately 71% classification accuracy and the highest macro F1-score. Nevertheless, all models struggled to distinguish metaphor from metonymy in many difficult cases, suggesting that the current feature representation is insufficient for perfect discrimination.

Future work should investigate contextual embeddings obtained from large language models together with richer semantic features that explicitly model different types of semantic shifts.